Table 14 Evaluation results on Matbench-Discovery [59], MDR phonon [47], elastic tensor [16, 35], and AdsorbML benchmarks [43]. Results are also provided for a diverse set of molecule [45] and molecular crystal [19, 30] benchmarks. NVE MD [23] tests whether energy conservation is observed when running the model for molecular dynamics. SOTA results from literature are reported where available. For the materials evaluations, UMA models were fine-tuned on MPtrj [17] and sAlex [5, 62] to be consistent with the benchmark's DFT settings.

	Materials									*Catalysis*	*Molecules*					*Molecular Crystals*		
Model	Matbench [59] *F1*	*RMSD*	*MAE [eV/atom]*	κ_{srme} [56]	Phonons [47] ω_{max} *[K]*	*Free Energy [kJ/mol]*	Elasticity [16, 35] G_{vrh} *[GPa]*	K_{vrh} *[GPa]*	NVE MD [23] *Conserve*	AdsorbML [43] *Success Rate*	OMol25 [45] *Ligand-strain [meV]*	*PDB-pocket [meV]*	*Dist-SR [meV]*	*Dist-LR [meV]*	NVE MD [23] *Conserve*	CSP Targets [30] *Latice Energy [kJ/mol]*	*Kendall Rank*	*RMSD [Å]*
UMA																		
UMA-S	0.916	0.064	0.020	0.203	17.59	5.0	8.54	4.96	✓	68.35%	4.39	150.3	67.6	432.1	✓	2.70	0.82	0.12
UMA-S-1	0.914	0.064	0.020	0.231	18.65	5.51	13.72	5.19	✓	64.85%	5.19	138.3	23.8	-	✓	2.57	0.81	0.13
UMA-S-1.1	0.913	0.064	0.020	0.204	18.82	5.48	9.47	5.16	✓	66.80%	5.2	127.7	26.7	256.7	✓	2.13	0.86	0.13
UMA-M-1.1	0.929	0.061	0.018	0.176	14.81	3.87	8.57	4.78	✓	72.25%	2.3	76.8	21.5	214.8	✓	3.24	0.82	0.14
Literature																		
eSEN-30M-OAM [23]	0.925	0.061	0.018	0.170	15.00	4.00	9.13	5.73	✓	-	-	-	-	-	-	-	-	-
ORB v3 [58]	0.905	0.075	0.024	0.210	-	-	-	-	-	-	-	-	-	-	-	-	-	-
SevenNet-MF-ompa [37]	0.901	0.064	0.021	0.317	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GRACE-2L-OAM [8]	0.880	0.067	0.023	0.294	-	-	-	-	-	-	-	-	-	-	-	-	-	-
MACE-MPA-0 [6]	0.852	0.073	0.028	0.412	-	-	-	-	-	-	-	-	-	-	-	-	-	-
eqv2-OC20 [43]	-	-	-	-	-	-	-	-	-	60.80%	-	-	-	-	-	-	-	-
GemNet-OC20 [43]	-	-	-	-	-	-	-	-	-	54.88%	-	-	-	-	-	-	-	-
eSEN-sm-cons. [45]	-	-	-	-	-	-	-	-	-	-	4.52	147.3	28.6	268.6	✓	-	-	-
ST Baselines																		
eSEN-S-OMol	-	-	-	-	-	-	-	-	-	-	5.15	154.48	73.02	608.9	✓	-	-	-
eSEN-S-OMC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	6.18	0.74	0.18

Table 15 Materials validation and test evaluations from OMat24 [5] and HEA. All energies are in meV, forces are in meV/Å and stresses are in meV/Å^3.

	Val [5]				*Test*											
					WBM [5]			OOD Composition [5]			OOD Element [5]			HEA		
Model	Energy/Atom	Forces	Stress	Force Cosine	Energy/Atom	Forces	Stress	Energy/Atom	Forces	Stress	Energy/Atom	Forces	Stress	Energy/Atom	Forces	Stress
UMA																
UMA-S	11.3	57.1	2.9	0.98	20.0	60.8	4.4	11.5	57.0	3.0	9.9	56.9	2.6	22.0	72.8	3.1
UMA-M	10.0	47.3	2.7	0.99	18.1	51.4	4.3	10.2	47.6	2.9	8.5	47.1	2.4	19.0	62.2	3.2
UMA-L	9.7	43.5	2.5	0.99	17.6	45.5	3.8	9.8	43.6	2.6	8.1	43.6	2.3	24.8	48.3	2.8
Literature																
eSEN-30M-OMat [23]	10.7	47.3	2.6	0.99	16.2	49.6	4.1	10.7	47.3	2.8	9.0	47.2	2.3	20.0	59.5	3.2
eqV2-86M-OMat [46]	10.0	44.9	2.4	0.99	14.9	46.3	3.6	10.0	44.5	2.5	8.8	44.7	2.1	20.3	47.0	2.7

Table 16 Materials evals results. We note that UMA models are fine-tuned on the MPTrj [17] and sAlex [5, 62] datasets to be consistent with the DFT settings of the benchmarks.

	Matbench [59]							*Kappa 103* [56]		*MDR Phonons* [47]					*Elasticity* [16, 35]		*Binary Elasticity*		*NVE MD* [23]
Model	F1	DAF	Precision	Accuracy	MAE *[eV/atom]*	RMSE *[eV/atom]*	r^2	κ_{sre}	κ_{srme}	ω_{max}, MAE *[K]*	ω_{avg}, MAE *[K]*	Entropy, MAE *[kJ/K/mol]*	C_v, MAE *[kJ/K/mol]*	Free Energy, MAE *[kJ/mol]*	G_{vrh}, MAE *[GPa]*	K_{vrh}, MAE *[GPa]*	G_{vrh}, MAE *[GPa]*	K_{vrh}, MAE *[GPa]*	Conservative
UMA																			
UMA-S	0.92	6.00	0.92	0.97	0.02	0.07	0.87	0.09	0.20	17.59	7.41	13.59	3.49	5.00	8.54	4.96	8.57	5.25	✓
UMA-M	0.93	6.08	0.93	0.98	0.02	0.07	0.87	0.09	0.20	13.91	5.11	9.63	2.66	3.39	8.40	4.76	7.07	4.75	✓
UMA-L	0.93	6.09	0.93	0.98	0.02	0.07	0.86	0.45	0.67	78.50	27.68	43.04	15.85	18.20	20.56	14.48	21.95	17.01	✗
Literature																			
eSEN-30M-OAM [23]	0.93	6.07	0.93	0.98	0.02	0.07	0.87	-	0.17	15.00	10.21	10.00	3.00	4.00	9.13	5.73	9.02	5.73	✓
eqV2-86M-OAM [5]	0.92	6.05	0.92	0.98	0.02	0.07	0.85	1.82	1.94	840.33	377.96	426.79	102.72	251.14	19.60	26.25	22.02	26.50	✗

In Table 16 we show full results for Matbench discovery [59], MDR phonon, elastic tensors, high entropy alloy IS2RE and NVE MD conservation benchmarks. The Matbench-Discovery benchmark evaluates a model's ability to predict ground-state thermodynamic stability by optimizing geometry and predicting energy. The thermal conductivity prediction task demands accurate modeling of harmonic and anharmonic phonons, which are crucial for precise predictions of thermal transport. The MDR Phonon benchmark assesses a model's performance in predicting phonon and vibrational thermodynamic properties. The MP elastic constant benchmark tests a model's accuracy in predicting bulk and shear moduli, requiring precise calculations of stress tensors and their derivatives with respect to cell deformations.

E.3 Catalysis

For catalysis, we show full validation and test results for OC20 [11] in Table 17. The structures in the dataset contain molecules, called adsorbates, interacting with surfaces. The goal is to estimate the adsorption energy, which is the change in energy as the adsorbates come into contact with the surface, and the forces on the atoms. Force MAEs are comparable across UMA-M and UMA-L to other state-of-the-art models. Adsorption energies for UMA are calculated by subtracting two total energy calculations as described in the main text, which improves results over prior models.

In addition to energy and forces MAEs, we use AdsorbML to evaluate the performance on the practically relevant task of finding the global minima adsorption energy [43]. One limitation of the originally proposed AdsorbML pipeline is that it requires a DFT evaluation of the ML-identified global minima structure. To make this benchmark more accessible to those without access to DFT, we propose a slightly altered version of this benchmark that only requires ML. Here, we follow the same AdsorbML pipeline but allow the use of the ML-predicted global minima energy, but success is now only considered if the energy is within 0.1 eV of the DFT minima. Previously, success did not enforce a lower bound so long a DFT evaluation confirmed that energy prediction is realized. Without DFT, we also set a lower bound of 0.1 eV to provide some flexibility to finding a better global minima than DFT. We show that metrics are still highly correlated when you perform the original evaluation to the one proposed here.

Table 17 Catalysis validation and test results on OC20 [11] metrics. All energies are in meV and forces are in meV/Å.

	Val (Total Energy)						*Test (Ads. Energy)*			
	ID			OOD-Both			ID		OOD-Both	
Model	Energy	Forces	Force Cosine	Energy	Forces	Force Cosine	Energy	Force	Energy	Force
UMA										
UMA-S	63.6	24.1	0.63	107.0	29.2	0.65	52.1	24.3	70.2	30.9
UMA-M	43.1	15.8	0.73	70.0	19.2	0.75	33.4	16.0	46.5	21.0
UMA-L	32.6	12.0	0.77	49.8	14.5	0.79	32.4	12.2	43.5	15.9
Literature										
eqV2-OC20 [46]	-	-	-	-	-	-	149.1	11.63	306.5	15.74
GemNet-OC20 [24]	-	-	-	-	-	-	163.5	16.33	343.3	23.11

E.4 Molecules

Table 18 Open Molecule 2025 [45] validation evaluations across biomolecules, electrolytes, metal complexes, neutral organics and OOD-comp. All energies are in meV and forces are in meV/Å. All models are trained with preview OMol25.

	Biomolecules		*Electrolytes*		*Metal Complexes*		*Neutral Organics*		*OOD-Comp*	
Model	Energy/Atom	Force	Energy/Atom	Force	Energy/Atom	Force	Energy/Atom	Force	Energy/Atom	Force
UMA										
UMA-S	0.53	5.69	2.69	11.65	4.63	37.85	1.00	13.15	3.62	12.02
UMA-M	0.44	3.95	2.28	10.21	3.60	28.81	0.68	7.00	3.21	9.90
UMA-L	0.33	2.90	1.13	4.52	3.35	24.85	0.65	5.02	2.39	5.83
Baseline										
UMA-S-OMol	0.54	6.06	2.52	12.63	4.27	37.30	0.84	13.00	3.69	12.78

We report results on molecules following the Open Molecules 2025 [45]. These include energy and force estimates for validation and test splits, as well as, numerous other tasks. The validation and test results are

Table 19 Open Molecule 2025 [45] test evaluations across biomolecules, electrolytes, metal complexes, neutral organics and OOD-comp, metal ligand, PDB-TM, reactivity, COD and anions. All energies are in meV and forces are in meV/Å. All models are trained with preview OMol25.

	Biomolecules		*Electrolytes*		*Metal Complexes*		*Neutral Organics*		*OOD-Comp*		*Metal Ligand*		*PDB-TM*		*Reactivity*		*COD*		*Anions*	
Model	Energy/Atom	Force	Energy/Atom	Force	Energy/Atom	Force	Energy/Atom	Force	Energy/Atom	Force	Energy/Atom	Force	Energy/Atom	Force	Energy/Atom	Force	Energy/Atom	Force	Energy/Atom	Force
UMA																				
UMA-S	0.51	5.70	3.80	13.95	3.07	33.56	1.49	20.33	3.64	10.80	1.23	17.71	0.88	16.12	4.82	61.80	2.92	29.82	0.66	10.85
UMA-M	0.42	3.89	3.29	13.19	2.40	24.85	0.98	10.83	3.26	9.09	0.99	12.04	0.69	10.37	3.88	47.75	2.19	20.11	0.50	8.03
UMA-L	0.34	2.92	1.41	5.42	2.47	21.67	1.03	6.91	2.33	5.19	1.05	10.00	0.81	8.76	3.93	41.97	2.57	15.44	0.62	5.90
Baseline																				
UMA-S-OMol	0.52	6.06	3.52	15.23	2.86	33.30	1.35	20.06	3.67	11.56	1.18	17.49	0.79	14.11	4.89	61.16	2.93	24.12	0.47	10.38

shown in Tables 18 and 19, respectively. Note that the UMA-S-OMol model is only trained on the preview OMol25 dataset, which is ≈ 70% of the full OMol25 dataset for a fair comparison with the UMA models that were trained on the same subset. In general, the UMA-S and UMA-S-OMol models provide comparable results.

Table 20 Open Molecule 2025 [45] single point evaluations for protein-ligand, IE/EA, spin gap and distance scaling. All energies are in meV and forces are in meV/Å. All models are trained with preview OMol25.

	Protein-ligand		*IE/EA*			*Spin gap*			*Distance scaling*			
Model	Ixn Energy *MAE*	Ixn Forces *MAE*	Δ Energy *MAE*	Δ Forces *MAE*	Δ Forces *cosine sim.*	Δ Energy *MAE*	Δ Forces *MAE*	Δ Forces *cosine sim.*	Δ Energy (SR) *MAE*	Δ Forces (SR) *MAE*	Δ Energy (LR) *MAE*	Δ Forces (LR) *MAE*
UMA												
UMA-S	150.25	5.09	336.16	80.60	0.77	665.75	112.09	0.69	67.60	4.11	432.14	5.54
UMA-M	89.69	4.06	236.76	66.28	0.81	547.73	98.15	0.74	41.60	3.86	588.74	8.73
UMA-L	71.68	2.27	244.23	57.18	0.82	568.36	93.09	0.73	16.55	2.03	246.10	2.34
Baselines												
UMA-S-OMol	154.48	5.59	310.19	77.48	0.77	634.02	110.28	0.70	73.02	4.16	608.91	7.69

Table 21 Open Molecule 2025 [45] optimization evaluations including ligand-strain, conformer prediction, and protonation states. Results are reported across a variety of energy and structure based metrics for each task. All energies are in meV. All models are trained with preview OMol25.

	Ligand strain		*Conformers*					*Protonation*			
Model	Strain energy *MAE [meV]↓*	RMSD *min. [Å]↓*	RMSD *ensemble [Å]↓*	RMSD *boltz. [Å]↓*	Δ Energy *MAE [meV]↓*	RMSD *reopt.[Å]↓*	Δ Energy *reopt.[meV]↓*	RMSD *ensemble [Å]↓*	Δ Energy *MAE [meV]↓*	RMSD *reopt.[Å]↓*	Δ Energy *reopt.[meV]↓*
UMA											
UMA-S	4.39	0.28	0.06	0.05	5.76	0.02	3.06	0.13	40.42	0.04	18.35
UMA-M	2.45	0.19	0.04	0.03	3.19	0.01	1.47	0.08	24.08	0.02	10.99
UMA-L	3.37	0.25	0.05	0.05	4.97	0.01	2.27	0.11	30.31	0.02	12.54
Baselines											
UMA-S-OMol	5.15	0.31	0.06	0.05	5.25	0.03	3.24	0.13	32.82	0.04	18.65

OMol25 [45] provides numerous other tasks for evaluation. These include evaluations which only require the estimation of a single point DFT calculation and evaluations that require optimizations. The comparison for single point DFT calculations is shown in Table 20. Similar to the test results, the UMA-S and UMA-S-OMol models perform similarly with the UMA-M and UMA-L performing significantly better. Table 21 shows the results for tasks that require optimizations, which require repeated calls to the model to update atoms positions until a local energy minima is found. The small models report similar performance, and the UMA-M demonstrates the highest accuracies. It is likely that UMA-M outperforms UMA-L due to UMA-M being energy conserving and better behaved during optimization tasks.

E.5 Molecular Crystals

Table 22 Open Molecular Crystals 2025 [19] validation and test table. All energies are in meV, forces are in meV/Å and stresses are in meV/$Å^3$.

	Val				*Test*			
Model	Energy/Atom	Forces	Stress	Force Cosine	Energy/Atom	Forces	Stress	Force Cosine
UMA								
UMA-S	0.9	4.9	1.0	0.92	0.9	4.8	1.0	0.93
UMA-M	0.8	3.1	1.0	0.95	0.8	3.0	1.0	0.95
UMA-L	0.6	2.3	0.1	0.96	0.6	2.3	0.1	0.96
Baselines								
UMA-S-OMC	1.06	5.58	0.96	0.92	1.05	5.39	0.94	0.92

Table 23 Open Molecular Crystals 2025 [19] evaluation for polymorphs from the Structure Ranking Phase of CCDC 7th CSP Blind Test [30]. All lattice energies are per molecule basis in kJ/mol.

	CCDC 7th CSP Blind Test Polymorphs							
	Lattice Energy			Rank correlation		Structures		
Model	*MAE [kJ/mol]*	*RMSE [kJ/mol]*	r^2	*Kendall*	*Spearman*	*RMSD [Å]*	*RMSD sd [Å]*	*Match rate*
UMA								
UMA-S	2.69	3.67	0.73	0.82	0.93	0.12	0.07	0.99
UMA-M	2.66	3.71	0.60	0.81	0.91	0.13	0.07	0.99
UMA-L	2.49	3.70	0.81	0.84	0.95	0.12	0.07	1.00
Baselines								
UMA-S-OMC	6.18	7.38	0.07	0.74	0.87	0.18	0.08	0.91

Open Molecular Crystals (OMC25) [19] evaluates whether a model can predict the packing of molecule into crystal structures. This task requires the accurate estimation of inter-molecular forces. Results on validation and test splits are shown in Table 22. It is notable that all sizes of UMA models outperform the UMA-S-OMC model trained on only OMC25. This indicates that the other datasets, such as OMol25, provide useful complementary information for the task. One important and real-world task for molecular crystals is to predict the lowest energy packing, called a polymorph, for a molecule. Results for this task for a subset of molecular crystal polymorphs from the most recent 7th Crystal Structure Prediction (CSP) Blind Test [30] are shown in Table 23. The pymatgen's [53] StructureMatcher class with default settings is used to match DFT and UMA-relaxed polymorphs, and root mean square deviation (RMSD) is computed for matches. Similar to the test metrics, the UMA models outperform the UMA-S-OMC trained on only OMC25.

E.6 DAC

The results on OpenDAC [68] are shown in Table 24. The OpenDAC dataset contains Metal Organic Frameworks (MOFs) with CO_2 and water molecules. The goal is to estimate the change in energy in the presence with and without the CO_2 and water molecules. These adsorption energies are computed in the same manner as for catalysts. The use of total energies leads to significantly better adsorption energy estimates, similar to catalysis. The forces of UMA-M and UMA-L are similar to the SoTA eqV2-ODAC [68] model.

Table 24 OpenDAC [68] val and test table. All energies are in meV and forces are in meV/Å.

	Val (Total Energy)			*Test (Ads. Energy)*					
				ID			*OOD-L/T*		
Model	Energy	Forces	Force Cosine	Energy	Forces	Force Cosine	Energy	Forces	Force Cosine
UMA									
UMA-S	60.4	5.9	0.82	169.5	16.7	0.63	292.4	16.0	0.57
UMA-M	59.3	3.8	0.91	167.3	14.8	0.62	290.2	10.7	0.76
UMA-L	38.7	3.3	0.91	177.1	7.8	0.82	291.1	6.5	0.91
Literature									
eqV2-ODAC [68]	-	-	-	145.0	8.2	0.69	316.0	7.2	0.72

F Additional MoLE Analysis

F.1 Expert Analysis

Using a limited validation set consisting of 10,000 OMat24, 5,000 OC20, 20,000 OMol25, 10,000 OMC25, and 5,000 ODAC23 samples, we calculate the mean expert coefficient for each element-expert pair across all systems where the pair appears. We visualize these results using 32 periodic tables, each representing one of the 32 experts in UMA-S (Figure 5).

Additionally we visualize the expert cofficient mean and variance across datasets (tasks). Many experts utilized in OC20 are also used in OMat24. In contrast, the experts associated with OMol25 show minimal overlap with other datasets, aside from a small subset shared with OM25C. Lastly ODAC23 and OMC25 utilize the fewest experts, and these two share a single expert between them.6).

F.2 Generalization Across Architectures

Table 25 Testing the generalization capability of MoLE layers. We applied 8-expert MoLE layers to both eSEN[23] and EquiformerV2[46] model architectures and measured their relative performance on direct pretraining against the versions without no MoLE. For each row, we show the relative improvement over the baseline.

OMol25 Val subset	eSEN 6M	EquiformerV2 6M
Metal Complexes	+10%	+10%
Spice	+5%	+5%
Neutral organics	0%	+3%
Biomolecules	+25%	+21%
Electrolytes	+12%	+10%
Mean across subsets	**+10%**	**+10%**

We performed additional analysis to ablate the benefit of the MoLE layers. Since our MoLE implement is very general and can be applied any linear layers inside a model, we ran experiments adding MoLE to both eSEN and EquiformersV2 architectures, controlling for model size (6M base parameters) and 8 training epochs on Omol dataset, direct pretraining only.

25 shows that despite EquiformersV2 and eSEN being very different architectures, the relative benefit of adding MoLE is fairly consistent in both cases. This suggests that MoLE is a simple and general approach to boost the capacity of MLIP models without incurring inference speed and memory overhead.